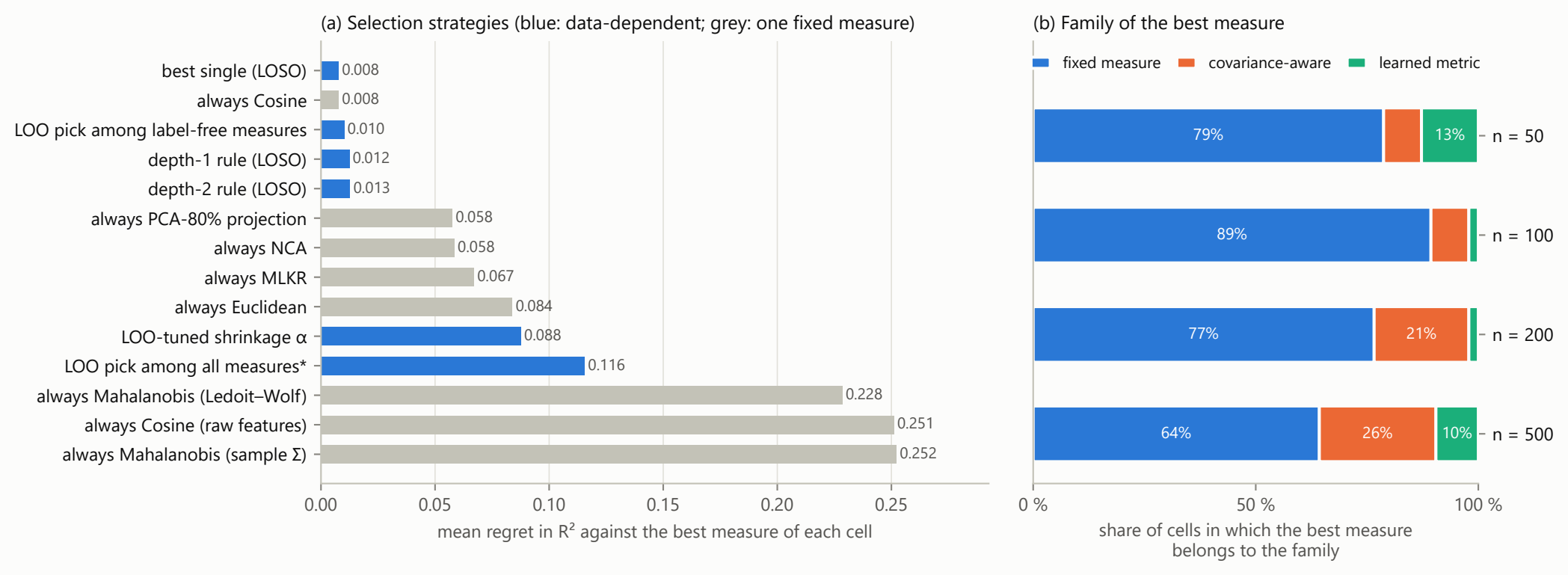
(a) Selection strategies (blue: data-dependent; grey: one fixed measure)
best single (LOSO)
always Cosine
LOO pick among label-free measures
depth-1 rule (LOSO)
depth-2 rule (LOSO)
always PCA-80% projection
always NCA
always MLKR
always Euclidean
LOO-tuned shrinkage α
LOO pick among all measures*
always Mahalanobis (Ledoit–Wolf)
always Cosine (raw features)
always Mahalanobis (sample Σ)
0.008
0.008
0.010
0.012
0.013
0.058
0.058
0.067
0.084
0.088
0.116
0.228
0.251
0.252
0.00
0.05
0.10
0.15
0.20
0.25
mean regret in R² against the best measure of each cell
(b) Family of the best measure
fixed measure
covariance-aware
learned metric
79%
13%
n = 50
89%
n = 100
77%
21%
n = 200
64%
26%
10%
n = 500
0 %
50 %
100 %
share of cells in which the best measure belongs to the family